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MICROCONTROLLER INCLUDING A DIGITAL POTENTIOMETER

Abstract

An apparatus comprises a microcontroller including a digital potentiometer provided in a microcontroller packaging. The digital potentiometer includes a resistor ladder including a plurality of resistors arranged in series and a plurality of selectively controllable potentiometer switches, a first resistor terminal connected to the resistor ladder, and a wiper terminal connected to the resistor ladder. The microcontroller includes a processor to transmit a wiper select signal to the digital potentiometer to selectively control the potentiometer switches to selectively connect at least a subset of the resistors between the first resistor terminal and wiper terminal, to thereby control the resistance between the first resistor terminal and wiper terminal. The microcontroller includes a first pin connected to the first resistor terminal and a second pin connected to the wiper terminal, the first and second pins being exposed through the microcontroller packaging to allow connection of the digital potentiometer to an external device.

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Background/Summary

RELATED APPLICATION [0001] This application claims priority to commonly owned U.S. Provisional Patent Application No. 63/561,473 filed Mar. 5, 2024, the entire contents of which are hereby incorporated by reference for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to digital potentiometers (DigiPots), and more particularly to a microcontroller including an integrated DigiPot.

BACKGROUND

[0003] A digital potentiometer or “DigiPot” is a digitally controlled device to produce a desired impedance (e.g., to adjust a voltage or current) to thereby provide a similar function as a mechanical potentiometer, a rheostat, or other device. DigiPots typically have an architecture including a “resistor ladder” or a digital-to-analog converter (DAC).

[0004] A resistor ladder type DigiPot includes (a) a plurality of resistors connected in series, called a “resistor ladder,” connected between a pair of resistor ladder terminals (often referred to as Terminals A and B) and (b) a wiper terminal (often referred to as Terminal W) that is selectively connected to a respective “step” along the resistor ladder using digital signals to control switches connected at different points along the resistor ladder. Such control signals are typically communicated over Serial Peripheral Interface (SPI) or Inter-Integrated-Circuit (I^{sup}.2C) interfaces.

[0005] The selected step on the resistor ladder determines the resistance ratio of the DigiPot. The number of steps is normally indicated with a bit value, for example where 8 bits provides 256 steps. Some example applications of DigiPots include sensor calibration, volume control, brightness control, and programmable filters.

[0006] Unlike a typical DAC, which uses a resistor ladder to produce a selected voltage and typically does not carry a current (or carries a very small current of less than 10 μ A), a typical DigiPot is designed to carry a current (e.g., 1-10 mA) and uses a resistor ladder to controllably limit the current, by producing a selected impedance.

[0007] A DigiPot may provide different functionality depending on the connection of the resistor ladder terminals (e.g., Terminals A and B). For example, a DigiPot having the resistor ladder terminals (e.g., Terminals A and B) connected to an electric circuit may function as a potentiometer. As another example, a DigiPot having one resistor ladder terminal connected to an electric circuit with the other resistor ladder terminal (e.g., Terminal B) floating may function as a rheostat. As another example, a DigiPot having one resistor ladder terminal (e.g., Terminal A) floating with the other resistor ladder terminal (e.g., Terminal B) connected to ground may function as a “grounded resistor” device. Accordingly, as used herein, a “digital potentiometer” or “DigiPot” may comprise a potentiometer, a rheostat, or a grounded resistor device, depending on the connection of the resistor ladder terminals.

[0008] Embedded microcontrollers commonly work with external stand-alone DigiPots, for example in the form of discrete potentiometer, rheostat, or grounded resistor packages. A stand-alone DigiPot package includes various pins connected to respective pins of an embedded microcontrollers (e.g., by wire bonds or solder bonds), to provide a supply voltage and control

signals to the stand-alone DigiPot.

[0009] There is a need to provide a microcontroller with an integrated DigiPot.

SUMMARY

[0010] Examples of the present disclosure provide a microcontroller (e.g., an embedded microcontroller) including a digital potentiometer (DigiPot) formed therein, referred to herein as an integrated DigiPot. The integrated DigiPot may comprise a potentiometer, a rheostat, or a grounded resistor device.

[0011] One aspect provides an apparatus comprising a microcontroller, the microcontroller comprising a microcontroller packaging and a digital potentiometer (DigiPot), a microcontroller memory, and a processor arranged in the microcontroller packaging. The DigiPot includes a resistor ladder including a plurality of resistors arranged in series and a plurality of potentiometer switches, a first resistor terminal connected to the resistor ladder, and a wiper terminal connected to the resistor ladder, wherein the plurality of potentiometer switches are selectively controllable to selectively connect at least a subset of the plurality of resistors between the first resistor terminal and the wiper terminal to define a resistance between the first resistor terminal and the wiper terminal. The microcontroller memory stores a wiper select signal. The processor to transmit the wiper select signal to the digital potentiometer to selectively control the plurality of switches to control the resistance between the first resistor terminal and the wiper terminal. The microcontroller includes a first pin connected or connectable to the first resistor terminal, and a second pin connected or connectable to the wiper terminal, wherein the first pin and the second pin are exposed through the microcontroller packaging to allow connection of the digital potentiometer to an electronic device external to the microcontroller.

[0012] In some examples, the microcontroller includes a charge pump to apply a charge pump voltage to the potentiometer switches. In some examples, the microcontroller includes switching circuitry to selectively switch between (a) a first switch state in which the supply voltage is applied to the potentiometer switches; and (b) a second switch state in which the charge pump voltage is applied to the potentiometer switches to maintain a constant impedance of the potentiometer switches, independent of variations in the supply voltage.

[0013] In some examples, the microcontroller includes a first pin switch connected between the first pin and the first resistor terminal, and a second pin switch connected between the second pin and the wiper terminal, and wherein the first pin switch and the second pin switch are controllable by the processor using respective pin switch control signals.

[0014] In some examples, the first pin switch is controllable to selectively connect the first pin to different selected circuitry, including at least (a) the first resistor terminal of the digital potentiometer and (b) a first other circuit component of the microcontroller; and the second pin switch is controllable to selectively connect the first pin to different selected circuitry, including at least (a) the second resistor terminal of the digital potentiometer and (b) a second other circuit component of the microcontroller.

[0015] In some examples, the microcontroller includes a charge pump to apply a constant charge pump voltage to the first pin switch and the second pin switch.

[0016] In some examples, the microcontroller include switching circuitry to selectively switch, based on a charge pump select signal, between (a) a first switch state in which a supply voltage is applied to the first pin switch and the second pin switch and (b) a second switch state in which the constant charge pump voltage is applied to the first pin switch and the second pin switch.

[0017] In some examples, the digital potentiometer comprises a potentiometer. In other examples, the digital potentiometer comprises a rheostat. In other examples, the digital potentiometer comprises a grounded resistor.

[0018] In some examples, the digital potentiometer comprises a second resistor terminal connected to the resistor ladder, wherein the plurality of potentiometer of resistors are connected between the first resistor terminal and the second resistor terminal; and the microcontroller comprises a third pin

connected to the second resistor terminal, wherein the third pin is exposed through the microcontroller packaging.

[0019] In some examples, the microcontroller includes a first pin switch connected between the first pin and the first resistor terminal, and a second pin switch connected between the second pin and the wiper terminal, and a third pin switch connected between the third pin and the second resistor terminal, wherein the first pin switch, the second pin switch, and the third pin switch are controllable by the processor using respective pin switch control signals.

[0020] One aspect provides an apparatus including an electronic device including electronic device circuitry, and a microcontroller connected to the electronic device. The microcontroller includes a microcontroller packaging, and a digital potentiometer arranged in the microcontroller packaging and including a resistor ladder including a plurality of resistors arranged in series, and a plurality of selectively controllable potentiometer switches, a first resistor terminal connected to the resistor ladder, and a wiper terminal connected to the resistor ladder. The microcontroller also includes a first pin connected to the first resistor terminal. And a second pin connected to the wiper terminal, wherein the first pin and the second pin are exposed through the microcontroller packaging and connectable to the electronic device. The microcontroller also includes a microcontroller memory arranged in the microcontroller packaging, the microcontroller memory storing a wiper select signal, and a processor arranged in the microcontroller packaging, the processor to transmit the wiper select signal to the digital potentiometer to selectively control the plurality of potentiometer switches to control the resistance between the first resistor terminal and the wiper terminal, and thereby selectively control a resistance through the electronic device circuitry of the electronic device.

[0021] In some examples of the apparatus, the microcontroller comprises a charge pump to apply a charge pump voltage to the potentiometer switches.

[0022] In some examples of the apparatus, the microcontroller comprises switching circuitry to selectively switch, based on a charge pump select signal, between (a) a first switch state in which the supply voltage is applied to the potentiometer switches; and (b) a second switch state in which the charge pump voltage is applied to the potentiometer switches to maintain a constant impedance of the potentiometer switches, independent of variations in the supply voltage.

[0023] In some examples, the microcontroller comprises a first pin switch connected between the first pin and the first resistor terminal, and a second pin switch connected between the second pin and the wiper terminal, wherein the first pin switch and the second pin switch are controllable by the processor using respective pin switch control signals.

[0024] In some examples, the first pin switch is controllable to selectively connect the first pin to either (a) the first resistor terminal of the digital potentiometer or (b) a first other circuit component of the microcontroller, and the second pin switch is controllable to selectively connect the second pin to either (a) the second resistor terminal of the digital potentiometer or (b) a second other circuit component of the microcontroller.

[0025] In some examples of the apparatus, the microcontroller comprises a charge pump to apply a constant charge pump voltage to the first pin switch and the second pin switch.

[0026] In some examples, the microcontroller comprises switching circuitry to selectively switch, based on a charge pump select signal, between (a) a first switch state in which a supply voltage is applied to the first pin switch and the second pin switch and (b) a second switch state in which the constant charge pump voltage is applied to the first pin switch and the second pin switch.

[0027] One aspect provides an apparatus comprising a microcontroller, the microcontroller comprising a microcontroller packaging and a digital potentiometer (DigiPot) arranged in the microcontroller packaging. The DigiPot includes a resistor ladder including a plurality of resistors arranged in series and a plurality of potentiometer switches, a first resistor terminal connected to the resistor ladder, and a wiper terminal connected to the resistor ladder, wherein the plurality of potentiometer switches are selectively controllable to selectively connect at least a subset of the

plurality of resistors between the first resistor terminal and the wiper terminal to define a resistance between the first resistor terminal and the wiper terminal. The microcontroller includes a charge pump to apply a charge pump voltage to respective potentiometer switches of the plurality of potentiometer switches.

[0028] In some examples, the microcontroller comprises switching circuitry to selectively switch between (a) a first switch state in which the supply voltage is applied to the potentiometer switches and (b) a second switch state in which the charge pump voltage is applied to the potentiometer switches.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] Example aspects of the present disclosure are described below in conjunction with the figures, in which:

[0030] FIG. 1 shows an example microcontroller including an integrated digital potentiometer (DigiPot);

[0031] FIG. 2 shows an example apparatus including the example microcontroller of FIG. 1 including the integrated DigiPot, and an example electronic device connected to the integrated DigiPot; and

[0032] FIG. 3 shows another example apparatus including another example microcontroller including an integrated DigiPot, and an example electronic device connected to the integrated DigiPot.

[0033] It should be understood that the reference number for any illustrated element that appears in multiple different figures has the same meaning across the multiple figures, and the mention or discussion herein of any illustrated element in the context of any particular figure also applies to each other figure, if any, in which that same illustrated element is shown.

DETAILED DESCRIPTION

[0034] Examples of the present disclosure provide a microcontroller (e.g., an embedded microcontroller) including a digital potentiometer (DigiPot) formed therein, referred to herein as an integrated DigiPot. The integrated DigiPot may be arranged as a potentiometer, a rheostat, or a grounded resistor device, e.g., depending on the respective implementation. The integrated DigiPot may provide various benefits. For example, the integrated DigiPot may reduce system cost, e.g., as compared with a stand-alone DigiPot connected to a microcontroller. For example, the microcontroller may include various components used by the DigiPot, for example non-volatile memory (NVM), thus allowing for a reduction in system components. As another example, the serial interface and control program included in a stand-alone DigiPot may be eliminated, as the DigiPot can be programmed by a processor (e.g., core) of the microcontroller, which may improve the speed or latency of control of the DigiPot. In addition, the additional package of the conventional stand-alone DigiPot may be eliminated. As another example, in some examples the integrated DigiPot may allow reduced test time and reduced test flows, e.g., as compared with a stand-alone DigiPot connected to a microcontroller.

[0035] FIG. 1 shows an example microcontroller **100** (e.g., an embedded microcontroller) including an example digital potentiometer (DigiPot) **102** integrated therein. The DigiPot **102** and various other components, including a microcontroller memory **104**, a processor **106**, a first pin **110**, a second pin **112**, and optionally a third pin **114** are provided in a microcontroller packaging **116**, e.g., comprising an epoxy mold compound (EMC) or other encapsulant.

[0036] The DigiPot **102** may include a resistor ladder **120** including a plurality of resistors **122** arranged in series, a plurality of potentiometer switches **124**, and a number of resistor terminals connected to the resistor ladder **120**, including a first resistor terminal **130**, a wiper terminal **132**,

and optionally a second resistor terminal **134**. The plurality of potentiometer switches **124** are selectively controllable to selectively connect at least a subset of the resistors **122** (i.e., a subset or all of the resistors **122**) between the first resistor terminal **130** and the wiper terminal **132** to define a resistance between the first resistor terminal **130** and the wiper terminal **132**. In some examples, the potentiometer switches **124** may be selectively controllable to bypass the resistor ladder **120**; in such examples the “at least a subset of the resistors **122**” selectively connected between the first resistor terminal **130** and the wiper terminal **132** may include none, some, or all of the resistors **122**, depending on the relevant control of the respective potentiometer switches **124**.

[0037] In contrast, the full plurality (series) of resistors **122** remains connected between the first resistor terminal **130** and the optional second resistor terminal **134**.

[0038] In some examples, the first resistor terminal **130** may be connected at one end (e.g., a “top” end) the resistor ladder **120**, and the optional second resistor terminal **134** may be connected at an opposite end (e.g., a “bottom” end) the resistor ladder **120**. In some examples, the first resistor terminal **130** may be referred to as “Terminal A,” the optional second resistor terminal **134** may be referred to as “Terminal B,” and the wiper terminal **132** may be referred to as “Terminal W.”

[0039] The microcontroller memory **104** may include any one or more data storage device, for example processor registers (e.g., digital control registers), an Electrically Erasable Programmable Read-Only Memory (EEPROM) device, and/or other non-volatile memory (NVM) device. The microcontroller memory **104** may store at least one wiper select signal **140** for selectively controlling the potentiometer switches **124** to control the resistance between the first resistor terminal **130** and the wiper terminal **132**.

[0040] The processor **106** may comprise a Central Processing Unit (CPU) including one or more processor cores, a Digital Signal Processors (DSP), Application-Specific Integrated Circuit (ASIC), and/or other type(s) or processor. The processor **106** may transmit the wiper select signal(s) **140** to the digital potentiometer **102** to selectively control respective ones of the potentiometer switches **124** to connect a selected subset of resistors **122** between the first resistor terminal **130** and wiper terminal **132**, to thereby control a resistance between the first resistor terminal **130** and wiper terminal **132**. In one example, the processor **106** may transmit the wiper select signal(s) **140** to the digital potentiometer **102** by writing to a register (not shown) of the digital potentiometer **102** via an internal data bus of microcontroller **100**.

[0041] The first resistor terminal **130** is connected or connectable to the first pin **110**, the wiper terminal **132** is connected or connectable to the second pin **112**, and the optional second resistor terminal **134** is connected or connectable to the optional third pin **114**. The first pin **110**, second pin **112**, and optional third pin **114** may be exposed through the microcontroller packaging **116** to allow connection of the DigiPot **102** to an external electronic device **150** (i.e., separate from the microcontroller **100**) via the first pin **110**, second pin **112**, and/or optional third pin **114**.

[0042] The term pin, as used herein, for example in the context of the first pin **110**, second pin **112**, and optional third pin **114**, includes any type of conductive contact, for example gull-wing or J-lead terminals, solder balls, or lands, without limitation.

[0043] In some examples, e.g., as shown in FIG. 3 discussed below, the first resistor terminal **130** is selectively connectable to the first pin **110** by controlling a first pin switch, the wiper terminal **132** is selectively connectable to the second pin **112** by controlling a second pin switch, and the optional second resistor terminal **134** is selectively connectable to the optional third pin **114** by controlling a third pin switch. As discussed below, these controllable pin switches may allow an external device (e.g., the example electronic device **150**) to selective connect to either the DigiPot **102** or other electronics in the microcontroller **100** via the first pin **110**, second pin **112**, and/or optional third pin **114**. In another example, the first resistor terminal **130** and wiper terminal **132** may be permanently connected to the first pin **110**, second pin **112**, and/or optional third pin **114**, respectively.

[0044] The potentiometer switches **124** carry current (e.g., in the range of milliamps) during operation, and thus inherently have an impedance, referred to herein as switch impedance, which is

exponentially proportional to voltage. In some examples, the microcontroller supply voltage (Vdd) may vary over time. These variations in the supply voltage (Vdd), if applied to the DigiPot, may affect the switch impedance of potentiometer switches **124**, which in some instances may significantly affect the performance of the DigiPot **102**, e.g., by reducing the predictability and reliability of the DigiPot **102** output. Thus, in some examples, the example microcontroller **100** may optionally include a charge pump (not shown in FIG. **1**) to apply a constant charge pump voltage (Vdd_hvp) to the DigiPot **102** (rather than the supply voltage Vdd), which may function to maintain a constant switch impedance of potentiometer switches **124**, independent of variations in the microcontroller supply voltage Vdd. The microcontroller **100** may include switching circuitry (e.g., a multiplexer) to selectively switch, based on a charge pump select signal effected by the processor **106**, between (a) a first switch state in which the supply voltage Vdd is applied to DigiPot **102** and (b) a second switch state in which the constant charge pump voltage Vdd_hvp is applied to DigiPot **102**. Example features of such charge pump and switching circuitry are discussed below with reference to FIG. **3**.

[0045] As used herein, the term “constant” means varying by less than +10%, or in some implementations less than +5%, over time. For example, a constant charge pump voltage (Vdd_hvp) as discussed herein refers to a charge pump voltage that varies by less than +10%, or in some implementations less than +5%, over time. Similarly, a constant switch impedance as discussed herein refers to an impedance of a switch (e.g., a potentiometer switch or a pin switch discussed herein) that varies by less than +10%, or in some implementations less than +5%, over time.

[0046] FIG. **2** shows an example apparatus **200** including the example microcontroller **100** including the integrated DigiPot **102** as discussed above, and an example electronic device **150** connected to the integrated DigiPot **102**. The electronic device **150** may include electronic device circuitry **152**. The electronic device **150** may comprise, for example, a speaker, a passive filter, an active filter, or any other suitable type of electronic device.

[0047] As shown, the electronic device circuitry **152** is connected to the first pin **110** and second pin **112** of the microcontroller **100**. The electronic device circuitry **152** may also be optionally connected to the optional third pin **114**. The microcontroller processor **106** may use wiper select settings **364** stored in the microcontroller memory **104** to send wiper select signal(s) **140** to the integrated DigiPot **102** to selectively control respective potentiometer switches **124** to control the resistance between the first resistor terminal **130** and wiper terminal **132**, and thereby selectively control a resistance through the electronic device circuitry **152**. As discussed above, the microcontroller **100** may include a charge pump to apply a constant charge pump voltage Vdd_hvp is applied to DigiPot **102**, to prevent or reduce variations in switch impedance of potentiometer switches **124** that may otherwise result from variations in the microcontroller supply voltage Vdd.

[0048] FIG. **3** shows an example apparatus **300** including an example microcontroller **302** including the integrated DigiPot **102**, and an example electronic device **150** connected to the integrated DigiPot **102**. The electronic device **150** may include electronic device circuitry **152**. The example microcontroller **302** may represent an example implementation of the example microcontroller **102** that includes various additional features related to the integrated DigiPot **102**, for example including (a) pin switches to selectively connect or disconnect the pins **110**, **112**, and **114** to the DigiPot **102** and (b) a charge pump to apply a constant voltage to the pin switches and the potentiometer switches **124** of the DigiPot **102** to maintain a respective switch impedance of the pin switches and potentiometer switches **124** constant, e.g., independent of variations in a supply voltage (Vdd), which may provide predictable performance of the DigiPot **102**. It should be understood that any or all of the features of example microcontroller **302** discussed below may be similarly included in the example microcontroller **102** discussed above.

[0049] As shown in FIG. **3**, the microcontroller **302** includes the elements of the example microcontroller **100** discussed above, including the DigiPot **102**, microcontroller memory **104**,

processor **106**, first pin **110**, second pin **112**, and optional third pin **114**, provided in the microcontroller packaging **116**, e.g., comprising EMC or other encapsulant. The DigiPot **102** includes the resistor ladder **120** including the plurality of resistors **122** arranged in series, the plurality of potentiometer switches **124**, the first resistor terminal **130**, wiper terminal **132**, and optional second resistor terminal **134**.

[0050] The microcontroller **302** also includes a first pin switch **310** connected between the first pin **110** and the first resistor terminal **130**, a second pin switch **312** connected between the second pin **112** and the wiper terminal **132**, and a third pin switch **314** connected between the third pin **114** and the second resistor terminal **134**. The first pin switch **310**, second pin switch **312**, and third pin switch **314** are controllable by the processor **106** using respective pin switch control signals. As shown, the first pin switch **310**, second pin switch **312**, and third pin switch **314** are connected to respective level shifters (LS) that can control the first pin switch **310**, second pin switch **312**, and third pin switch **314** based on respective pin switch control signals. In particular, a first level shifter **320** connected to the first pin switch **310** may control the first pin switch **310** based on a first pin switch control signal **330** to selectively connect (or disconnect) the first pin **110** to the DigiPot **102**; a second level shifter **322** connected to the second pin switch **312** may control the second pin switch **312** based on a second pin switch control signal **332** to selectively connect (or disconnect) the second pin **112** to the DigiPot **102**; and a third level shifter **324** connected to the third pin switch **314** may control the third pin switch **314** based on a third pin switch control signal **334** to selectively connect (or disconnect) the third pin **114** to the DigiPot **102**.

[0051] The processor **106** may generate the pin switch control signals **330**, **332**, and **334**, along with a DigiPot enable signal **336** to enable/disable the DigiPot **102**, based on DigiPot enable settings **360** stored in the microcontroller memory **104**.

[0052] In some examples, the first pin switch **310**, second pin switch **312**, and third pin switch **314** are provided in an input selector multiplexer (MUX) for selectively (alternatively) connecting the first pin **110**, second pin **112**, and third pin **114** to multiple different functional circuits, including the DigiPot **102** and other functional circuits, thereby allowing the external electronic device **150** to selectively connect to the DigiPot **102** or to other functional circuitry of the microcontroller **302** using the same pins **110**, **112**, and **114**, e.g., depending on different needs or functioning of the electronic device **150** at different times.

[0053] The pin switches **310**, **312**, and **314** and potentiometer switches **124** in the DigiPot **102** carry current (e.g., in the range of milliamps) during operation, and thus inherently have an impedance, referred to herein as switch impedance. The switch impedance of a respective switches (e.g., pin switches **310**, **312**, and **314** and potentiometer switch **124**) is exponentially proportional to voltage. In one example, the switch impedance of a respective potentiometer switch **124** is about 5 k Ω at 5V. In some examples, the switch impedance may significantly affect the performance of the DigiPot **102**, especially wherein the switch impedance constitutes a significant fraction of the overall impedance of the DigiPot **102**, for example where the switch impedance is significant relative to the total string impedance of the resistor ladder **120**.

[0054] Because the switch impedance varies exponentially as a function of voltage, variations in V_{dd} applied to the pin switches **310**, **312**, and **314** and/or DigiPot **102** over time (e.g., depending on the current status of the microcontroller **302**) may significantly reduce the predictability and reliability of the output of the DigiPot **102**.

[0055] Thus, the example microcontroller **302** includes a charge pump **340** to apply a constant charge pump voltage V_{dd_hvp} to the pin switches **310**, **312**, and **314** and the DigiPot **102** (in particular, potentiometer switches **124**), e.g., independent of V_{dd} and changes in V_{dd} over time. As shown in FIG. 3, the charge pump **340** may supply the charge pump voltage V_{dd_hvp} to (a) respective level shifters **320**, **322**, and **324** provided at the respective pin switches **310**, **312**, and **314** pins to maintain the pin switches **310**, **312**, and **314** at the charge pump voltage V_{dd_hvp}, and thereby maintain a constant switch impedance of respective pin switches **310**, **312**, and **314**, and (b)

to the DigiPot **102** to maintain the potentiometer switches **124** at the charge pump voltage Vdd_hvp, and thereby maintain a constant switch impedance of respective potentiometer switches **124**.

[0056] In some examples, the microcontroller **302** may selectively enable/disable the charge pump **340** using a CP_enable signal **342** generated by a charge pump select setting **364** stored in memory **104**. In addition, the microcontroller **302** may selectively switch between the digital supply voltage (Vdd) or the charge pump voltage (Vdd_hvp) to be applied at the pin switches **310**, **312**, and **314**, for example depending on whether the DigiPot **102** is currently enabled. In the example shown in FIG. 3, the microcontroller **302** includes switching circuitry **346** to select between the digital supply voltage (Vdd) and charge pump voltage (Vdd_hvp), which selection may be controlled by the processor **106** using a Vdd_hvp_select signal **350**. In some examples, the switching circuitry **346** comprises a multiplexer (MUX).

[0057] The disclosed microcontrollers (e.g., embedded microcontrollers) including an integrated DigiPot may provide various advantages over conventional microcontroller connected to a stand-alone DigiPot. For example, various components and pins of the stand-alone DigiPot may be eliminated, as the respective circuitry may be provided internal to the microcontroller. For example, the non-volatile memory (NVM) of a stand-alone DigiPot may be eliminated. As another example, the serial interface of a stand-alone DigiPot may be eliminated, as the DigiPot may be programmed by the microcontroller processor (e.g., CPU). As another example, the packaging of the stand-alone DigiPot may be eliminated. Based on the above, costs may be significantly reduced as compared with conventional designs.

[0058] The example microcontrollers **100** and **302** shown in FIGS. 1-3 may be implemented as potentiometers, rheostats, or grounded resistors. For example, microcontroller **100** or **302** may be implemented as a potentiometer, wherein all three pins **110**, **112**, and **114** are connected to corresponding DigiPot terminals **130**, **132**, and **134**, respectively. As another example, microcontroller **100** or **302** may be implemented as a rheostat may, wherein pins **110** and **112** are connected to corresponding DigiPot terminals **130**, **132**, respectively, but wherein the second terminal **134** is allowed to float. As another example, microcontroller **100** or **302** may be implemented as a grounded resistor, wherein the second pin **112** is connected to the wiper terminal **132**, wherein the first terminal **130** is allowed to float, and wherein the second terminal **134** is tied to ground.

[0059] Although example embodiments have been described above, other variations and embodiments may be made from this disclosure without departing from the spirit and scope of these embodiments.

Claims

1. An apparatus, comprising: a microcontroller, comprising: a microcontroller packaging; a digital potentiometer provided in the microcontroller packaging and including: a resistor ladder including a plurality of resistors arranged in series and a plurality of potentiometer switches; a first resistor terminal connected to the resistor ladder; a wiper terminal connected to the resistor ladder; wherein the plurality of potentiometer switches are selectively controllable to selectively connect at least a subset of the plurality of resistors between the first resistor terminal and the wiper terminal to define a resistance between the first resistor terminal and the wiper terminal; a microcontroller memory arranged in the microcontroller packaging, the microcontroller memory storing a wiper select signal; a processor arranged in the microcontroller packaging, the processor to transmit the wiper select signal to the digital potentiometer to selectively control the plurality of switches to control the resistance between the first resistor terminal and the wiper terminal; a first pin connected or connectable to the first resistor terminal; and a second pin connected or connectable to the wiper terminal; wherein the first pin and the second pin are exposed through the

microcontroller packaging to allow connection of the digital potentiometer to an electronic device external to the microcontroller.

2. The apparatus of claim 1, wherein the microcontroller comprises a charge pump to apply a charge pump voltage to the potentiometer switches.

3. The apparatus of claim 2, wherein the microcontroller comprises switching circuitry to selectively switch between: (a) a first switch state in which the supply voltage is applied to the potentiometer switches and (b) a second switch state in which the charge pump voltage is applied to the potentiometer switches to maintain a constant impedance of the potentiometer switches, independent of variations in the supply voltage.

4. The apparatus of claim 1, wherein the microcontroller comprises: a first pin switch connected between the first pin and the first resistor terminal; and a second pin switch connected between the second pin and the wiper terminal; and wherein the first pin switch and the second pin switch are controllable by the processor using respective pin switch control signals.

5. The apparatus of claim 4, wherein: the first pin switch is controllable to selectively connect the first pin to different selected circuitry, including at least (a) the first resistor terminal of the digital potentiometer and (b) a first other circuit component of the microcontroller; and the second pin switch is controllable to selectively connect the first pin to different selected circuitry, including at least (a) the second resistor terminal of the digital potentiometer and (b) a second other circuit component of the microcontroller.

6. The apparatus of claim 4, wherein the microcontroller comprises a charge pump to apply a constant charge pump voltage to the first pin switch and the second pin switch.

7. The apparatus of claim 4, wherein the microcontroller comprises switching circuitry to selectively switch, based on a charge pump select signal, between (a) a first switch state in which the supply voltage is applied to the first pin switch and the second pin switch and (b) a second switch state in which the constant charge pump voltage is applied to the first pin switch and the second pin switch.

8. The apparatus of claim 1, wherein the digital potentiometer comprises a potentiometer.

9. The apparatus of claim 1, wherein the digital potentiometer comprises a rheostat.

10. The apparatus of claim 1, wherein: the digital potentiometer comprises a second resistor terminal connected to the resistor ladder, wherein the plurality of potentiometer of resistors are connected between the first resistor terminal and the second resistor terminal; and the microcontroller comprises a third pin connected to the second resistor terminal, wherein the third pin is exposed through the microcontroller packaging.

11. The apparatus of claim 10, comprising: a first pin switch connected between the first pin and the first resistor terminal; a second pin switch connected between the second pin and the wiper terminal; and a third pin switch connected between the third pin and the second resistor terminal; wherein the first pin switch, the second pin switch, and the third pin switch are controllable by the processor using respective pin switch control signals.

12. An apparatus, comprising: an electronic device including electronic device circuitry; and a microcontroller connected to the electronic device, the microcontroller comprising: a microcontroller packaging; a digital potentiometer arranged in the microcontroller packaging and including: a resistor ladder including a plurality of resistors arranged in series, and a plurality of selectively controllable potentiometer switches; a first resistor terminal connected to the resistor ladder; a wiper terminal connected to the resistor ladder; a first pin connected to the first resistor terminal; and a second pin connected to the wiper terminal; wherein the first pin and the second pin are exposed through the microcontroller packaging and connectable to the electronic device; a microcontroller memory arranged in the microcontroller packaging, the microcontroller memory storing a wiper select signal; and a processor arranged in the microcontroller packaging, the processor to transmit the wiper select signal to the digital potentiometer to selectively control the plurality of potentiometer switches to control the resistance between the first resistor terminal and

the wiper terminal, and thereby selectively control a resistance through the electronic device circuitry of the electronic device.

13. The apparatus of claim 12, wherein the microcontroller comprises a charge pump to apply a charge pump voltage to the potentiometer switches.

14. The apparatus of claim 12, wherein the microcontroller comprises switching circuitry to selectively switch, based on a charge pump select signal, between: (a) a first switch state in which the supply voltage is applied to the potentiometer switches and (b) a second switch state in which the charge pump voltage is applied to the potentiometer switches to maintain a constant impedance of the potentiometer switches, independent of variations in the supply voltage.

15. The apparatus of claim 12, wherein the microcontroller comprises: a first pin switch connected between the first pin and the first resistor terminal; and a second pin switch connected between the second pin and the wiper terminal; and wherein the first pin switch and the second pin switch are controllable by the processor using respective pin switch control signals.

16. The apparatus of claim 15, wherein: the first pin switch is controllable to selectively connect the first pin to either (a) the first resistor terminal of the digital potentiometer or (b) a first other circuit component of the microcontroller; and the second pin switch is controllable to selectively connect the second pin to either (a) the second resistor terminal of the digital potentiometer or (b) a second other circuit component of the microcontroller.

17. The apparatus of claim 15, wherein the microcontroller comprises a charge pump to apply a constant charge pump voltage to the first pin switch and the second pin switch.

18. The apparatus of claim 17, wherein the microcontroller comprises switching circuitry to selectively switch, based on a charge pump select signal, between (a) a first switch state in which the supply voltage is applied to the first pin switch and the second pin switch and (b) a second switch state in which the constant charge pump voltage is applied to the first pin switch and the second pin switch.

19. An apparatus, comprising: a microcontroller, comprising: a microcontroller packaging; a digital potentiometer provided in the microcontroller packaging and including: a resistor ladder including a plurality of resistors arranged in series and a plurality of potentiometer switches; a first resistor terminal connected to the resistor ladder; a wiper terminal connected to the resistor ladder; wherein the plurality of potentiometer switches are selectively controllable to selectively connect at least a subset of the plurality of resistors between the first resistor terminal and the wiper terminal to define a resistance between the first resistor terminal and the wiper terminal; and a charge pump to apply a charge pump voltage to respective potentiometer switches of the plurality of potentiometer switches.

20. The apparatus of claim 19, the microcontroller comprises switching circuitry to selectively switch between (a) a first switch state in which the supply voltage is applied to the potentiometer switches and (b) a second switch state in which the charge pump voltage is applied to the potentiometer switches.
